EXP 801 DS Wheat Heat Drought

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Li-6800 STOMATAL CONDUCTANCE MODEL PARAMETER ESTIMATION - LIGHT ON TO LIGHT OFF (DECREASE)

- Using the final model fitted to the data, obtain the parameters of interest
- These include:
- k_d, being the time constant for gs to increase to Gsmax
- lambda, being the lag in the response time of gs to a step increase in light
- g0, being the starting value of the gs response (intercept/upper asymptote)
- g1, being the lower asymptote of the gs response
- Slmax, a derived trait from McAusland paper

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.li6800.stomatalCondParamEstimation_d.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/decrease/")
library(dae)
library(dplyr)
library(brms) ##package used to fit bayesian regression model using Stan
library(tidyr)
library(emmeans)
library(tidybayes)
```

Set up constants and files

```
options(width=90)
load("../data/stomatalCondModel_d.v2.Rdata")
saveFile <- "../data/stomatalCondParamEstimation_d.v1.Rdata"
```

Have a look at the final model summary and store fixed effects

```
summary(fit_off_final)
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + Room_Temperature * Watering + (1 | Rep)
         g1 ~ 1 + Room_Temperature * Watering + (1 | Rep)
         k ~ 1 + Watering
         lambda ~ 1 + Watering
   Data: li6800.off.dat (Number of observations: 1451)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000

Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.07      0.04     0.02     0.16 1.00    11785    14141
```

```
sd(g1_Intercept)     0.02      0.01     0.01     0.03 1.00    13170    14993

Regression Coefficients:
                                        Estimate Est.Error l-95% CI u-95% CI Rhat
g0_Intercept                                0.27      0.04     0.20     0.36 1.00
g0_Room_TemperatureCER8_Amb                -0.07      0.01    -0.09    -0.05 1.00
g0_WateringUnlim                            0.18      0.01     0.16     0.20 1.00
g0_Room_TemperatureCER8_Amb:WateringUnlim   -0.10      0.01    -0.13    -0.08 1.00
g1_Intercept                                0.00      0.01    -0.01     0.02 1.00
g1_Room_TemperatureCER8_Amb                -0.00      0.00    -0.01     0.00 1.00
g1_WateringUnlim                            0.08      0.00     0.07     0.08 1.00
g1_Room_TemperatureCER8_Amb:WateringUnlim   -0.06      0.00    -0.07    -0.06 1.00
k_Intercept                                 3.07      0.08     2.92     3.22 1.00
k_WateringUnlim                             3.50      0.09     3.33     3.67 1.00
lambda_Intercept                            0.19      0.13     0.01     0.49 1.00
lambda_WateringUnlim                        0.98      0.22     0.54     1.41 1.00
                                        Bulk_ESS Tail_ESS
g0_Intercept                               10754    12041
g0_Room_TemperatureCER8_Amb                17643    17443
g0_WateringUnlim                           17217    17981
g0_Room_TemperatureCER8_Amb:WateringUnlim  16292    17108
g1_Intercept                               11775    13342
g1_Room_TemperatureCER8_Amb                24306    19896
g1_WateringUnlim                           22213    19406
g1_Room_TemperatureCER8_Amb:WateringUnlim  20295    18877
k_Intercept                                26763    17678
k_WateringUnlim                            27430    16888
lambda_Intercept                           16580    10020
lambda_WateringUnlim                       23583    14106

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.03      0.00     0.03     0.03 1.00    31230    17133

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Build a dataframe of model parameters

```
li6800.d.param <- expand.grid(Room_Temperature = levels(li6800.off.dat$Room_Temperature),
                              Watering = levels(li6800.off.dat$Watering))
```

Use the emmeans package to get out the expected parameter estimates, along with HPD (credible intervals)

```
g0.emmeans <- as.data.frame(emmeans(fit_off_final,
                                    ~ Room_Temperature * Watering,
                                    nlpar = "g0"))
g0.emmeans <- g0.emmeans %>% rename_with(~paste0("g0_", .x), -all_of(c("Room_Temperature", "Watering")))
g0.emmeans <- within(g0.emmeans, {
  g0_tabquant <- paste0(round(g0_emmean, 3), " [", round(g0_lower.HPD, 3), ", ", round(g0_upper.HPD, 3)
```

```
})
li6800.d.param <- left_join(li6800.d.param, g0.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Watering)`
```

```
g1.emmeans <- as.data.frame(emmeans(fit_off_final,
                                    ~ Room_Temperature * Watering,
                                    nlpar = "g1"))
g1.emmeans <- g1.emmeans %>% rename_with(~paste0("g1_", .x), -all_of(c("Room_Temperature", "Watering")))
g1.emmeans <- within(g1.emmeans, {
  g1_tabquant <- paste0(round(g1_emmean, 3), " [", round(g1_lower.HPD, 3), ", ", round(g1_upper.HPD, 3)

})
li6800.d.param <- left_join(li6800.d.param, g1.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Watering)`
```

```
k.emmeans <- as.data.frame(emmeans(fit_off_final,
                                    ~ Watering,
                                    nlpar = "k"))
k.emmeans <- k.emmeans %>% rename_with(~paste0("k_", .x), -all_of(c("Watering")))
k.emmeans <- within(k.emmeans, {
  k_tabquant <- paste0(round(k_emmean, 3), " [", round(k_lower.HPD, 3), ", ", round(k_upper.HPD, 3), "]

})
li6800.d.param <- left_join(li6800.d.param, k.emmeans)
```

```
Joining with `by = join_by(Watering)`
```

```
lambda.emmeans <- as.data.frame(emmeans(fit_off_final,
                                    ~ Watering,
                                    nlpar = "lambda"))
lambda.emmeans <- lambda.emmeans %>% rename_with(~paste0("lambda_", .x), -all_of(c("Watering")))
lambda.emmeans <- within(lambda.emmeans, {
  lambda_tabquant <- paste0(round(lambda_emmean, 3), " [", round(lambda_lower.HPD, 3), ", ", round(lambd

})
li6800.d.param <- left_join(li6800.d.param, lambda.emmeans)
```

```
Joining with `by = join_by(Watering)`
```

Using these parameters, calculate Slmax

- Slmax looks to be reported in the units mmol mˆ-2 sˆ-2
- This involves multiplying g1 and g0 by 1000 and also k by 60
- I can calculate this using the posterior draws for each parameter and then combine them together

```
g0_draws <- emmeans(fit_off_final,
                    ~ Room_Temperature * Watering,
                    nlpar = "g0") %>%
  gather_emmeans_draws()

g1_draws <- emmeans(fit_off_final,
                    ~ Room_Temperature * Watering,
                    nlpar = "g1") %>%
  gather_emmeans_draws()
```

```
k_draws <- emmeans(fit_off_final,
                   ~ Watering,
                   nlpar = "k") %>%
  gather_emmeans_draws()

combined_draws <- g0_draws %>%
  rename(g0 = .value) %>%
  left_join(g1_draws %>% rename(g1 = .value),
            by = c(".draw", "Room_Temperature", "Watering")) %>%
  left_join(k_draws %>% rename(k = .value),
            by = c(".draw", "Watering"))

combined_draws <- combined_draws %>%
  mutate(Slmax = (g1*1000 - g0*1000) / (exp(1) * k*60))

summary_slopes <- combined_draws %>%
  group_by(Room_Temperature, Watering) %>%
  median_hdi(Slmax) %>%
  rename(Slmax_emmean = Slmax) %>%
  rename(Slmax_lower.HPD = .lower) %>%
  rename(Slmax_upper.HPD = .upper)

summary_slopes <- within(summary_slopes, {
  Slmax_tabquant <- paste0(round(Slmax_emmean, 3), " [", round(Slmax_lower.HPD, 3), ", ", round(Slmax_u

})

li6800.d.param <- left_join(li6800.d.param,
                            summary_slopes %>% select(Room_Temperature,
                                                      Watering,
                                                      Slmax_emmean,
                                                      Slmax_lower.HPD,
                                                      Slmax_upper.HPD,
                                                      Slmax_tabquant))
```

```
Joining with `by = join_by(Room_Temperature, Watering)`
```

```
li6800.d.param
```

```
  Room_Temperature Watering g0_emmean g0_lower.HPD g0_upper.HPD          g0_tabquant
1        CER7_Heat      Lim 0.2707174    0.1963175    0.3565208 0.271 [0.196, 0.357]
2         CER8_Amb      Lim 0.2002530    0.1229866    0.2836654   0.2 [0.123, 0.284]
3        CER7_Heat    Unlim 0.4552246    0.3812559    0.5407923 0.455 [0.381, 0.541]
4         CER8_Amb    Unlim 0.2838894    0.2080176    0.3661285 0.284 [0.208, 0.366]
     g1_emmean g1_lower.HPD g1_upper.HPD           g1_tabquant k_emmean k_lower.HPD
1 0.0038352005 -0.014231691   0.02066783 0.004 [-0.014, 0.021] 3.068592    2.916036
2 0.0009570562 -0.017151334   0.01810996 0.001 [-0.017, 0.018] 3.068592    2.916036
3 0.0791044184  0.061177916   0.09662226  0.079 [0.061, 0.097] 6.569226    6.362446
4 0.0132003079 -0.004716062   0.03086419 0.013 [-0.005, 0.031] 6.569226    6.362446
  k_upper.HPD           k_tabquant lambda_emmean lambda_lower.HPD lambda_upper.HPD
1    3.222151 3.069 [2.916, 3.222]     0.1691432     3.195929e-05        0.4367896
2    3.222151 3.069 [2.916, 3.222]     0.1691432     3.195929e-05        0.4367896
3    6.776745 6.569 [6.362, 6.777]     1.1651820     6.885239e-01        1.6543842
4    6.776745 6.569 [6.362, 6.777]     1.1651820     6.885239e-01        1.6543842
```

```
       lambda_tabquant Slmax_emmean Slmax_lower.HPD Slmax_upper.HPD
1     0.169 [0, 0.437]   -0.5344196      -0.7114045      -0.3762393
2     0.169 [0, 0.437]   -0.3991272      -0.5745690      -0.2422679
3 1.165 [0.689, 1.654]   -0.3513450      -0.4339158      -0.2806673
4 1.165 [0.689, 1.654]   -0.2529170      -0.3335238      -0.1809828
           Slmax_tabquant
1 -0.534 [-0.711, -0.376]
2 -0.399 [-0.575, -0.242]
3 -0.351 [-0.434, -0.281]
4 -0.253 [-0.334, -0.181]
```

Format this in a nicer way for export to Excel

```
li6800.d.export <- li6800.d.param %>% select(Room_Temperature, Watering,
                                             grep("tabquant", colnames(.))) %>%
  pivot_longer(cols = 3:ncol(.),
               names_to = "params",
               values_to = "values") %>%
  pivot_wider(names_from = c("Room_Temperature", "Watering"),
              values_from = "values")
li6800.d.export
```

```
# A tibble: 5 × 5
  params          CER7_Heat_Lim           CER8_Amb_Lim      CER7_Heat_Unlim CER8_Amb_Unlim
  <chr>           <chr>                   <chr>             <chr>           <chr>
1 g0_tabquant     0.271 [0.196, 0.357]    0.2 [0.123, 0.28... 0.455 [0.381, ... 0.284 [0.208,...
2 g1_tabquant     0.004 [-0.014, 0.021]   0.001 [-0.017, 0... 0.079 [0.061, ... 0.013 [-0.005...
3 k_tabquant      3.069 [2.916, 3.222]    3.069 [2.916, 3.... 6.569 [6.362, ... 6.569 [6.362,...
4 lambda_tabquant 0.169 [0, 0.437]        0.169 [0, 0.437]  1.165 [0.689, ... 1.165 [0.689,...
5 Slmax_tabquant  -0.534 [-0.711, -0.376] -0.399 [-0.575, ... -0.351 [-0.434... -0.253 [-0.33...
```

Save data and export

```
save.image(saveFile)
write.csv(li6800.d.export, "../data/li6800.d.param.export.csv", row.names = FALSE)
```